Hanyuan Xiao

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Introduction

PhD candidate at USC ICT specializing in 3D vision, neural rendering, and generative AI systems. Research focuses on robust multiview reconstruction, 3D foundation models, neural human capture, and production-oriented ML pipelines with quantified speed, memory, and robustness improvements.

Education

- **University of Southern California (USC)** — Los Angeles, CA
  Doctorate (PhD) in Computer Science; GPA: 3.88 — *September 2019 — August 2026*
  - **Honor**: Annenberg Fellowship
- **University of Southern California (USC)** — Los Angeles, CA
  Master of Science (M.S.) in Computer Science; GPA: 3.88 — *September 2019 — August 2025*
- **Rensselaer Polytechnic Institute (RPI)** — Troy, NY
  Bachelor of Science (B.S.) in Computer Science and Electrical Engineering (dual major); GPA: 3.95 — *September 2015 — May 2019*
  - **Honor**: Dean's Honor List (2015-2018), Distinguished Student

Work Experience

- **AI Scholar** — Seattle, WA
  Axon Inc. — *May 2025 — Present*
  - **Simulation Platform**: Built a CARLA/Unreal simulation framework for law-enforcement traffic and vehicle scenarios, supporting repeatable corner-case generation and model validation.
  - **Unreal Engine Tools**: Implemented Blueprint/C++ tools for vehicle physics, movement control, camera control, and scenario authoring in CARLA.
  - **ML Validation**: Integrated simulation with ML models via RTSP on Azure, exposing real-time metrics for safety-relevant corner cases and delivering 500+ synthetic videos for finetuning and robustness analysis.
- **Research Intern** — San Jose, CA
  Sony — *May 2021 — December 2021*
  - **Neural Capture**: Developed a sparse-view RGBA neural capture system for clothed human subjects using a parametric body prior and deformable neural volume representation.
  - **Training Pipeline**: Implemented data preprocessing, multiview sampling, and rendering-based supervision for end-to-end training and inference on real capture data.
  - **Patent**: Contributed to a U.S. patent on deformable clothing reconstruction within a neural volume conditioned on a parametric human body model.
- **Electric Energy System Designer** — Nanjing, China
  Jiangsu Longtu Zhaorun Engineering Design Co. Ltd — *July 2017 — August 2017*

Selected Research Projects

- **3D Generative Models for 3D Reconstruction, Editing and Generation**
  - **Image Covisibility Pre-grouping for 3D Foundation Models.** Built a multiview image disambiguation module that pre-groups covisible views for 3D foundation model inference; achieved **37% speedup** over prior methods and reduces time complexity from **O(n^2)** pairwise matching to **O(1)** per-query routing for large multiview sets, while improving accuracy on ambiguous/near-duplicate views.
  - **3D-Aware Video Diffusion Model (in progress).** Deriving a novel denoising process for video diffusion models to improve multiview consistency of high-frequency details; finetuning (post-training) 3D foundation model (VGGT) to synthesize frames from different bandwidths on a collection of large-scale video datasets.
  - **Text-Guided Scene Editing.** Built a one-shot 3D scene editing system driven by text prompts with illumination consistency; achieved **70.51% user preference** over prior work for photorealistic edits and developed a **GUI tool** to support rapid iteration, demos, and downstream applications.
- **Neural Human Performance Capture**
  - **Real-Time Volumetric Human Capture from Sparse Views.** Implemented a high-FPS neural capture system from sparse multiview inputs using sparse CNN features within a NeRF-style volume; achieved **O(1)** growth in inference time and VRAM w.r.t. rendering resolution, and delivers **32× GPU memory reduction** vs. prior state-of-the-art at the same resolution.

- **High-Resolution Volumetric Rendering.** Enabled high-resolution volumes up to 800^3; achieved **46× memory reduction** compared to dense sampling-based rendering while maintaining visual quality.
- **MVS-PERF: SMPL-X + Clothing Performance Capture.** Designed a multiview pipeline to learn deformable clothing conditioned on parametric human body models and reconstruct clothed neural volumes; contribution resulted in a **US patent**.

- **Physically-based Neural Rendering**
  - **Reflection-based NeRF with component decomposition.** Co-developed a neural rendering pipeline that models lighting (including subsurface scattering), material, and light transport in a neural volume; achieved **40–50× speedup** vs. **Autodesk Maya** at matched resolution with comparable rendering quality.
  - **High-frequency detail rendering.** Enabled high-frequency normal embedding to support rendering under arbitrary resolutions; improved rendering quality up to 36.69 **in PSNR**.

Publications & Patents

Under Review

- **Hanyuan Xiao**, Gonglin Chen, Haolin Xiong, Wenbin Teng, Haiwei Chen, Wei Su, Zhengping Ji, Victor Zhu, Yajie Zhao, "MVDG : Efficient Multi-view 3D Disambiguation on Unconstrained Real-World Images"

Publications

- Gonglin Chen, Ben Southall, **Hanyuan Xiao**, Wenbin Teng, Haolin Xiong, Tianwen Fu, Junyi Ouyang, Kshitij Singh Minhas, Supun Samarasekera, Rakesh Kumar, Yajie Zhao , "XDG: Accelerated Visual Disambiguation with Geometric 3D Features", *WACV 2027*
- Haolin Xiong, Tianwen Fu, Pratusha Bhuvana Prasad, Yunxuan Cai, Wenbin Teng, **Hanyuan Xiao**, Haiwei Chen, Yajie Zhao, "Mind-to-Face: Neural-Driven Photorealistic Avatar Synthesis via EEG Decoding", *ECCV 2026 (Oral)*
- Gonglin Chen, Tianwen Fu, Haiwei Chen, Wenbin Teng, **Hanyuan Xiao**, Yajie Zhao, "RDD: Robust Feature Detector and Descriptor using Deformable Transformer", *CVPR 2025*
- Haolin Xiong, Sairisheek Muttukuru, **Hanyuan Xiao**, Rishi Upadhyay, Pradyumna Chari, Yajie Zhao, Achuta Kadambi, "SparseGS: Real-Time 360° Sparse View Synthesis using Gaussian Splatting", *3DV 2025*
- **Hanyuan Xiao**, Yingshu Chen, Huajian Huang, Haolin Xiong, Jing Yang, Pratusha Prasad, Yajie Zhao, "Localized Gaussian Splatting Editing with Contextual Awareness", *WACV 2025*
- **Jing Yang*, Hanyuan Xiao**[*1], Wenbin Teng, Yunxuan Cai, Yajie Zhao, "Light Sampling Field and BRDF Representation for Physically-Based Neural Rendering", *ICLR 2023*

Patents

- Qing Zhang, **Hanyuan Xiao**, "Multiview Neural Human Prediction Using Implicit Differentiable Renderer For Facial Expression, Body", *US11961266B2*
- Yajie Zhao, Jing Yang, **Hanyuan Xiao**, "Systems and Methods for Physically-based Neural Face Shader via Volumetric Lightmaps", *US12002150B2*

Skills

- **Programming:** Python, C++, C, HTML, Bash
- **ML / Vision:** PyTorch, TensorFlow, NumPy, scikit-learn, OpenCV, Transformers, diffusion models, NeRF, 3DGS
- **Graphics / Simulation:** Blender, Autodesk Maya, Unreal Engine, CARLA, OpenGL, WebGL
- **Systems / Tools:** Git, Linux, Azure, Docker, RTSP

Academic Service

- **Conference Reviewer:** CVPR, ICCV, ECCV, NeurIPS, WACV, 3DV

[1] equal contribution